GenTL Producer for MATLAB® usage

October 2026
Rev 2026.1.3

20283 State Road 7, Suite 350
Boca Raton, FL 33498
USA
+1 (561) 689-2899

info@kaya.vision
support@kaya.vision

www.kaya.vision

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

Table of Contents

1 Introduction 1
1.1 Safety precautions 1
1.2 Disclaimer 2
2 **Overview** 3
3 **Installation** 4
3.1 Matlab R2025b 4
4 **Image Acquisition tool** 6
5 **Example** 7
REFERENCES 9

List of Figures

Figure 1 – MATLAB installation screen 4
Figure 2 – MATLAB APPS menu - Image Acquisition Explorer 6
Figure 3 – Image Acquisition Explorer 6

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

1 Introduction

1.1 Safety precautions

Please take the time to read through the precautions listed below in order to prevent preventable and unnecessary injuries and damage to you, other personnel or property. Read these safety instructions carefully prior to your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual in order to prevent misuse of product.

⚠ Caution! Read Carefully and do not disregard these instructions.
In the event of a failure, disconnect the power supply Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.
If an unpleasant smell or smoking occurs, disconnect the power supply. Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.
Do not disassemble, repair or modify the product. This may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel prior to inspection, modification or repair.
Do not place the product on unstable surfaces. Otherwise, it may drop or fall, resulting in injury to persons or the camera.
Do not use the product if dropped or damaged. Otherwise, a fire or electric shock may occur.
Do not touch the product with metallic objects. Otherwise, a fire or electric shock may occur.
Do not place the product in dusty or humid environments, nor where water may splash. Otherwise, a fire or electric shock may occur.
Do not wet the product or touch it with wet hands. Otherwise, the product may fail or it may cause a fire, smoking or electric shock.
Do not touch the gold-plated sections of the connectors on the product. Otherwise, the surface of the connector may be contaminated by sweat or skin-oil, resulting in contact failure of a connector, malfunction, fire or electric shock due to static electricity discharge.
Do not use or place the product in the following locations. ▪ Unventilated areas such as closets or bookshelves. ▪ Near oils, smoke or steam. ▪ Next to heat sources. ▪ A closed (and not running) car where the temperature becomes high. ▪ Static electricity replete locations ▪ Near water or chemicals. Otherwise, a fire, electric shock, accident or deformation may occur due to a short circuit or heat generation.
Do not place heavy objects on the product. Otherwise, the product may be damaged.
Be sure to discharge static electricity from body before touching any sensitive electronic components. The electronic circuits in your computer and the circuits on the camera are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

1.2 Disclaimer

This product should only be used for image capturing and processing. KAYA Vision assumes no responsibility for any damage that may ensue by using the camera for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company takes no responsibility for any damages caused by:

- Earthquake, thunder strike, natural disasters, a fire caused by usage beyond our control, willful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions
- Secondary damages caused by the use of this product or its unusable state (business interruption or others)
- Use of this product in any manner that contradicts this manual or malfunctions due to connection to other devices.
- Damage to this product that is out of our control or failure due to modification.
- Accidents and/or third parties that may be involved.

Additionally, KAYA Vision assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product
- Any consequences or other abnormalities following the use of this product

Repairs to this product are carried out by replacing it on a chargeable basis and not by repairing the faulty device. Non-chargeable replacement is offered for the initial failure, as long as it is reported no later than two weeks post-delivery of the product.

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

2 Overview

The MATLAB® product family provides a flexible environment for solving complex imaging problems in a wide range of applications including machine vision.
Image Acquisition Toolbox™ enables to acquire images and video from cameras and frame grabbers directly into MATLAB® and Simulink®. Image Processing Toolbox™ and Computer Vision Toolbox™ provide algorithms and tools for image processing, analysis, visualization and algorithm development.

GenTL standard defines a generic programming interface standard for machine vision cameras.
MATLAB® and Simulink® users are able to integrate standard GenTL compliant cameras/frame grabbers into their workflows to capture live video and images for processing.

3 Installation

3.1 Matlab R2025b

To use a Gen<I>Cam-compatible camera with MATLAB R2025b, install MATLAB, the KAYA Vision Point II SDK package containing the GenTL Producer, and the Image Acquisition Toolbox Support Package for Gen<I>Cam Interface.

1. **Install MATLAB R2025b**

Install MATLAB R2025b on the host computer.
Make sure that Image Acquisition Toolbox™ is installed and available under the MATLAB license. Image Acquisition Toolbox™ provides the interface for acquiring images from Gen<I>Cam GenTL hardware.

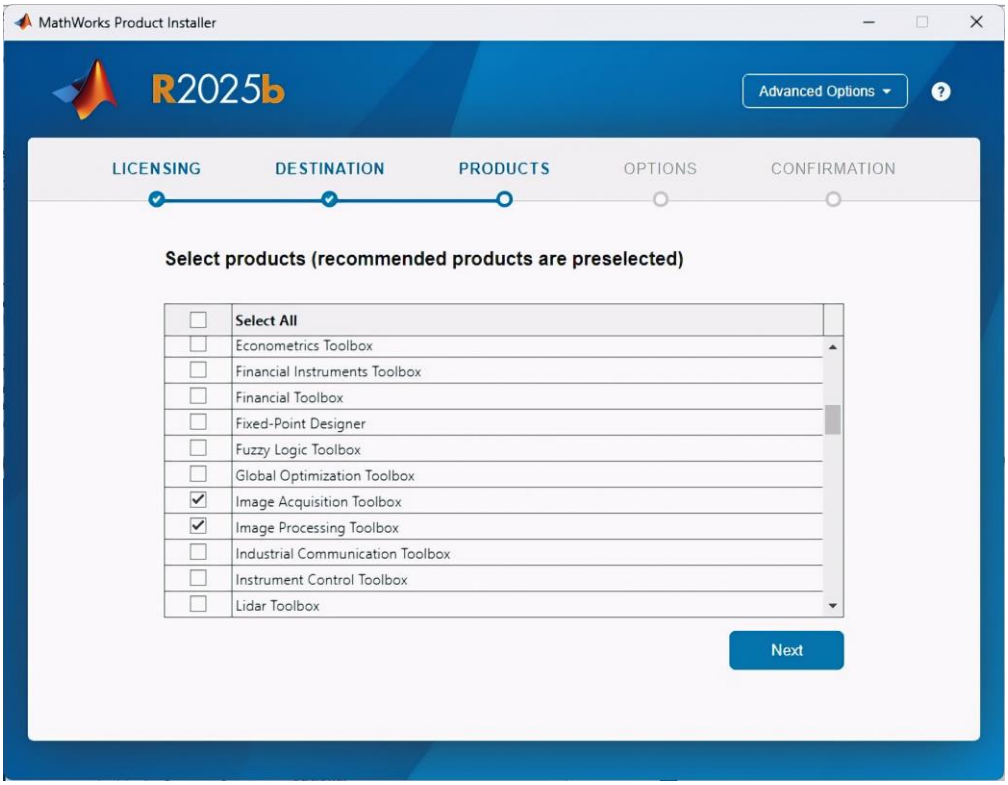


Figure 1 – MATLAB installation screen

2. **Install the Vision Point II SDK package**.

Install the Vision Point II SDK package which contains the KAYA Instruments KYVPGenTL_vc141.cti GenTL Producer.

3. **Install the Gen<I>Cam Interface support package.**

1) Open MATLAB R2025b. On the Home tab, select Add-Ons > Get Hardware Support Packages.
2) In Add-On Explorer, locate: **Image Acquisition Toolbox Support Package for Gen<I>Cam Interface.** This support package provides the MATLAB files required to use GenTL hardware with the GenTL adaptor.
3) Select the support package and follow the installation instructions displayed by MATLAB.
4) If prompted, sign in to the MathWorks Account.
5) Review and accept the applicable license agreements.
6) Complete the installation.
7) The support package is installed through MATLAB Add-On Explorer.

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

4. Verify the GenTL adaptor.

After installation, open the MATLAB Command Window and enter: `imaqhwinfo`
The returned list of installed adaptors should include: `gentl`
The gentl adaptor is the MATLAB interface used to access Gen<I>Cam GenTL cameras.

Check the available GenTL devices with: `imaqhwinfo("gentl")`
If the KAYA GenTL Producer is correctly installed and configured, the connected Gen<I>Cam camera should be listed.

5. Verify the GenTL Producer.

If the camera is not detected, run: `imaqsupport`
The GenTL section of the output provides information about the configured GenTL environment and detected GenTL Producers. MathWorks recommends using this command for troubleshooting Gen<I>Cam GenTL configuration.

The GenTL Producer must be correctly registered through the appropriate environment variable: `GENICAM_GENTL64_PATH`
The variable should point to the directory containing the GenTL Producer KYVPGenTL_vc141.cti file.

6. Restart MATLAB

After completing the installation and configuration, restart MATLAB R2025b.
The Gen<I>Cam camera can then be accessed through the GenTL adaptor.

4 Image Acquisition tool

The Image Acquisition Explorer is a graphical interface for connecting to image acquisition hardware, configuration acquisition parameters, live video previewing, and acquiring image and video data.
To open the app, select **Apps > Image Processing and Computer Vision > Image Acquisition Explorer**.
Alternatively, enter `imageAcquisitionExplorer` in the MATLAB Command Window.

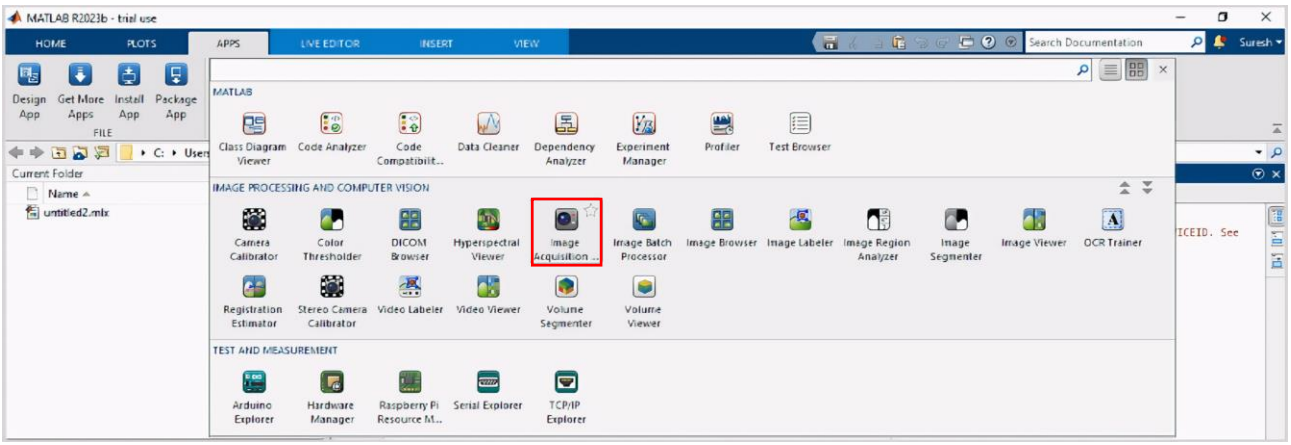


Figure 2 – MATLAB APPS menu - Image Acquisition Explorer

The preview panel displays the live image stream from the connected camera. Changes to camera properties and acquisition parameters are reflected in the preview in real time, allowing to verify the camera configuration before acquiring image data.

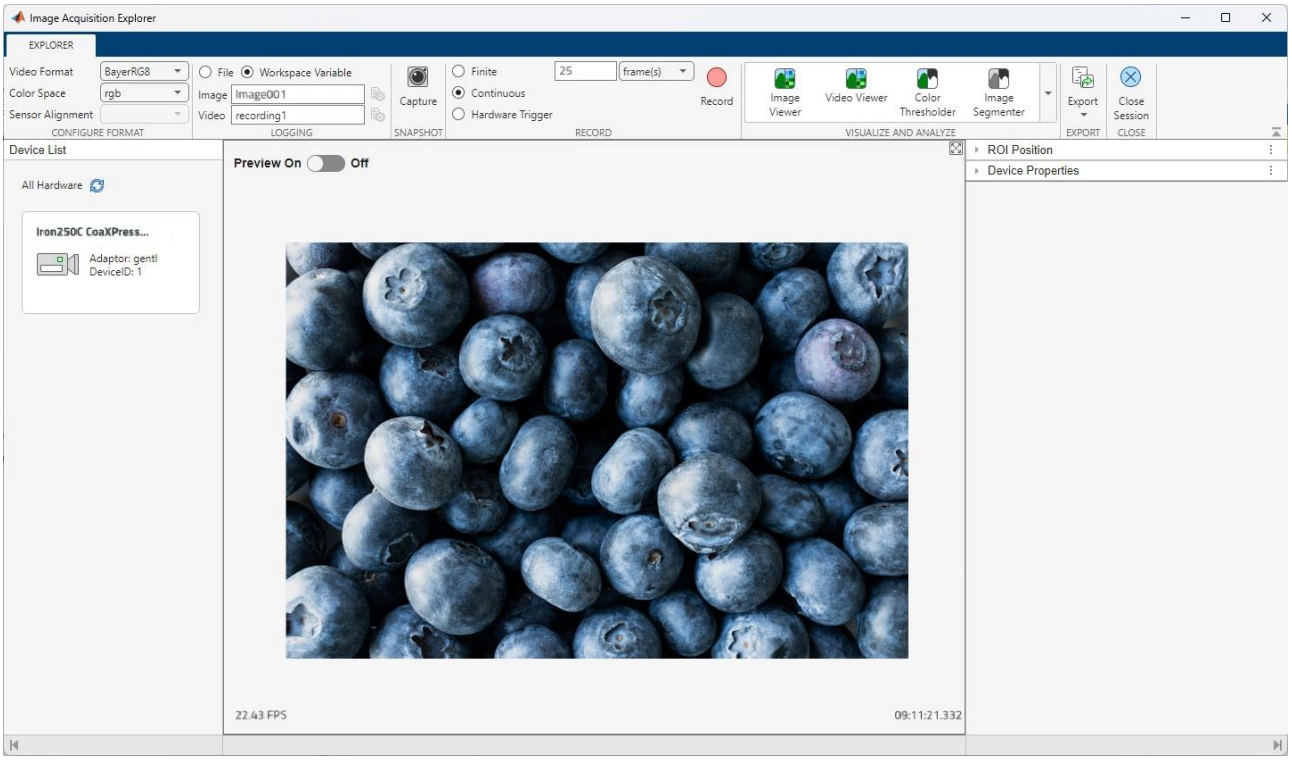


Figure 3 – Image Acquisition Explorer

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

5 Example

The toolbox has a comprehensive set of functions for command line programming of tasks such as device connection, image data acquisition, configuration of acquisition parameters and more.
The code below shows how to connect, configure and start frame acquisition. Acquired data can be retrieved using the created snapshot matrix:

```
% Access an image acquisition device

>> vidobj = videoinput('gentl', 1, 'Mono8');

% List the video source object's properties and their current values

>> get(vidobj)

% To access a specific property value, use the get function with the object and property
name

>> videoFormat = get(vidobj,'VideoFormat')

% List the video input object's configurable properties.

>> set(vidobj)

% To configure an object's property value, use the set function with the object, property
name, and property value or directly set a property value.

>> roivalue = [0, 0, 1024, 512]

   set(vidobj, 'ROIPosition', roivalue)

   vidobj.FramesPerTrigger = 50;

% Configure the object for manual trigger mode

>> triggerconfig(vidobj, 'manual');

% To obtain a property's description, use the imaqhelp function with the object and
property name. imaqhelp can also be used for function help.

>> imaqhelp(vidobj, 'LoggingMode')

% To obtain information on a property's attributes, use the propinfo function with the
object and property name.

>> propinfo(vidobj, 'LoggingMode')



% Acquire Multiple Frames

>> start(vidobj);                          % start acquisition

   for i = 1:100

        snapshot = getsnapshot(vidobj);    % get raw data of the acquired frame

        imagesc(snapshot);                 % show the acquired data as image

   end
```

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

```
   stop(vidobj);                          % stop acquisition

% Cleanup the image acquisition object and the MATLAB® workspace

>> delete(vidobj);

   clear vidobj;
```

For more examples and information visit: http://www.mathworks.com/help/imaq/examples.html

REFERENCES

Supported vision standards:

CoaXPress GEN<i>CAM

Vision Point documentation:

Vision Point

TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Vision support service:

- Create a support request on the web: support.kaya.vision
- Our knowledge base is available on: kb.kaya.vision

Visit us at www.kaya.vision for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model

KAYA VISION

Have questions about pricing, availability, documentation, or custom options?
We're always ready to assist and provide expert guidance.
Sales Inquiries: info@kaya.vision
Technical Support: support@kaya.vision
www.kaya.vision

KAYA Vision, Inc.
20283 State Road 7
Suite 350
Boca Raton, FL 33498
USA
+1 561 698-2899

2025 KAYA Vision, Inc. All rights reserved. KAYA Vision, the KAYA Vision Komodo logo, Predator, Iron, Zinc, Mercury and combinations thereof are trademarks of KAYA Vision, Inc. in the United States and/or other jurisdictions. Microsoft Windows® is a registered trademark of Microsoft Corporation. Linux® is a registered trademark of Linus Torvalds in the U.S. and other countries. JetPack® is a trademark of NVIDIA Corporation. HALCON® is a registered trademark of MVTec Software GmbH. Neither KAYA Vision, nor any of its products or services are affiliated with, endorsed by, or sponsored by National Instruments. MATLAB® is a registered trademark of The MathWorks, Inc. Cognex® is a registered trademark of Cognex Corporation. Other names are for informational purposes only and may be trademarks of their respective owners. KAYA Vision is not liable for harm or damage incurred by information contained in this document